Project Documentation

Car Parking Booking System

Submitted By:

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EICT_IITK Course Name: Full Stack Web Development with MERN Stack (MongoDB, Express, React, Node.js)

Abstract

This project is a Car Parking Booking System that enables users to find and book parking slots in real time. The system is designed with role-based access, where users can browse available slots and make bookings instantly, while administrators can manage locations, pricing, and availability. The application demonstrates CRUD operations, secure authentication, and a clean, intuitive user interface.

Introduction

Finding parking in busy urban areas is often time-consuming and frustrating. This project aims to simplify the process by providing a web-based solution where users can check availability, select a parking slot, and confirm their booking instantly. The system ensures transparency with per-minute pricing and provides admins with the ability to manage slots effectively.

Objectives

- Provide an easy way for users to book parking slots.
- Allow admins to add locations, manage slots, and set pricing.
- Ensure role-based access so that users only see their own bookings.
- Deliver a modern and responsive UI for a smooth experience.
- Demonstrate CRUD operations using MongoDB.

System Requirements

Software:

- Frontend: React.js
- Backend: Node.js + Express.js
- Database: MongoDB
- Other Tools: JWT for authentication, Context API for state management

Hardware:

- Minimum 4GB RAM
- Dual Core Processor
- Stable internet connection

System Design

Frontend (React): Handles user interface, routing, and API integration.
Backend (Express + MongoDB): Stores user data, bookings, and slot availability.
Authentication: JWT tokens secure login and role-based routing.

Modules / Features

User Module

- Login/Signup
- Browse locations
- Select parking slots from locations
- View bookings
- Manage own bookings

Admin Module

- Add parking locations
- Manage slots (availability)
- Update pricing
- View all user bookings for their location

Booking Module

- Real-time slot selection
- Transparent pricing
- Instant confirmation

Implementation Details

CRUD Operations:
- Create: Users create bookings; Admins create locations/slots.
- Read: View slots, bookings, pricing.
- Update: Change booking status, availability, and prices.
- Delete: Cancel bookings, remove slots.

UI/UX: Built with React, styled for clarity and modern aesthetics.

Future Scope

- Support for multiple admins with independent locations.
- Integration of online payment gateways.
- Maps API integration for selecting locations visually.
- Mobile app version for Android/iOS.

Conclusion

The Car Parking Booking System successfully demonstrates the use of React, Express, and MongoDB in building a full-stack application. It provides a practical, real-world solution while also fulfilling academic requirements such as role-based access, CRUD operations, and clean UI. This project strengthened skills in web development, database management, and UI/UX design.

References

- React.js Documentation
- MongoDB Documentation
- Express.js Documentation
- MDN Web Docs

Test Credentials

For evaluation, the following accounts are pre-created:

Admin Account

- Email: `admin@example.com`
- Password: `Admin@12345`

User Accounts

- User1 → Email: `user1@example.com`, Password: `user1@12345`
- User2 → Email: `user2@example.com`, Password: `user2@12345`
